

Chapter - Solutions (Complete Notes)

~ Complete Handwritten Chemistry Notes with Diagrams & Applications ~

Representation of concentration using symbols:

$\mathcal{A}$ = Solvent

$\mathcal{V}$ = Volume

$\mathcal{B}$ = Solute

n = moles

$\mathcal{W}$ = mass

$\mathcal{M}$ = molar mass

$$1. \text{ mass } \% = [\mathcal{W}_{\mathcal{B}} / (\mathcal{W}_{\mathcal{A}} + \mathcal{W}_{\mathcal{B}})] \times 100$$

$$2. \text{ Vol } \% = [\mathcal{V}_{\mathcal{B}} / (\mathcal{V}_{\mathcal{A}} + \mathcal{V}_{\mathcal{B}})] \times 100$$

$$3. \chi_{\mathcal{A}} = n_{\mathcal{A}} / (n_{\mathcal{A}} + n_{\mathcal{B}}) \quad | \quad \chi_{\mathcal{B}} = n_{\mathcal{B}} / (n_{\mathcal{A}} + n_{\mathcal{B}})$$
$$\chi_{\mathcal{A}} + \chi_{\mathcal{B}} = 1$$

$$4. \text{ ppm} = [\mathcal{W}_{\mathcal{B}} / (\mathcal{W}_{\mathcal{A}} + \mathcal{W}_{\mathcal{B}})] \times 10^6$$

$$5. \text{ Molarity} = n_{\mathcal{B}} / \mathcal{V}_{(\mathcal{A}+\mathcal{B})} = \mathcal{W}_{\mathcal{B}} / (\mathcal{M}_{\mathcal{B}} \times \mathcal{V}_{(\mathcal{A}+\mathcal{B})})$$

$$6. \text{ Molality} = n_{\mathcal{B}} / \mathcal{W}_{\mathcal{A}} = \mathcal{W}_{\mathcal{B}} / (\mathcal{M}_{\mathcal{B}} \times \mathcal{W}_{\mathcal{A}(\text{kg})})$$

Mixture:

When two or more chemically non-reactive substances are mixed they form a mixture.

Most of the materials we encounter in our daily lives are in the form of mixtures.

- i) Homogeneous mixture: It consists of single phase i.e., it is uniform and its properties are same throughout the mixture.
- ii) Heterogeneous mixture: It consists of distinct phases i.e., it is not uniform. It can be seen by naked eye.

Solution:

A homogeneous mixture of two or more components whose composition can be varied within certain limits is termed as a solution. We can't filtrate out the components of solution. All solutions are characterized by:

1. Homogeneity
2. Absence of settling
3. Particle size range is $(10^{-7} - 10^{-8} \text{ cm})$

Solvent & Solute:

Solvent: The component that is present in the largest quantity.

Solute: One or more components present in the solution other than solvent are called solutes.

Expressing Concentration of Solution:

- i) Mass percentage (W/W): It is defined as the amount of solute in grams in 100 g of the solution.

$$\text{Mass \%} = (\text{mass of solute} / \text{mass of solution}) \times 100$$

- ii) Volume percentage (V/V): It is defined as the volume of solute in ml present in 100 ml solution.

$$\text{Volume \%} = (\text{volume of solute} / \text{volume of solution}) \times 100$$

- iii) Mass by volume % (W/V): Defined as the mass of solute present in 100 ml of solution.

$$\text{Mass by volume \%} = (\text{mass of solute} / \text{volume of solution}) \times 100$$

- iv) Parts per million (ppm): When the solute is present in trace quantities, it is convenient to express the concentration in parts per million.

$$\text{ppm} = (\text{mass of solute} / \text{mass of solution}) \times 10^6 \text{ or } \text{mass \%} \times 10^4$$

- v) Mole fraction (χ): It is defined as the ratio of number of moles of one component to the total no. of moles of the solution.

$$\text{Mole fraction of solute } (\chi_B) = n_B / (n_A + n_B)$$

$$\text{Mole fraction of solvent } (\chi_A) = n_A / (n_A + n_B)$$

Mole fraction is unitless quantity. $\chi_A + \chi_B = 1$

- vi) Molarity (M): Molarity is defined as the number of moles of solute dissolved in one litre of solution. It is denoted by M .

$$M = (\text{no. of mole of solute} / \text{vol. of solution in litre}) \quad \text{Unit} = \text{mol/litre}$$

- vii) Molality (m): No. of moles of solute dissolved in 1 kg of solvent. It is denoted by " m ".

$$m = (\text{no. of mole of solute} / \text{mass of solvent in kg}) \quad \text{Unit} = \text{mol/kg}$$

Colligative Properties:

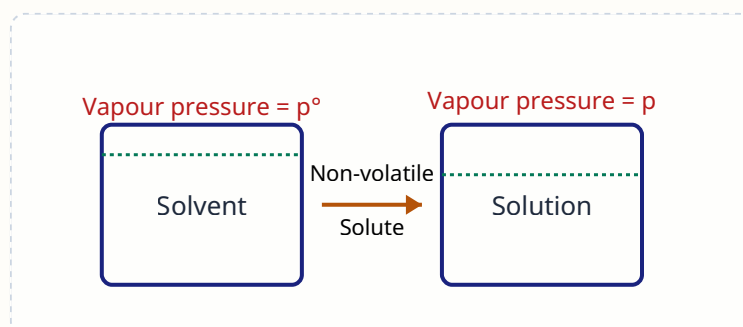
Properties depends only on the number of solute particles present in the solution irrespective of their nature. These properties are known as colligative properties.

1) Relative Lowering of Vapour Pressure (RLVP):

Relative lowering of vapour pressure, a colligative property, is the ratio of the reduction in vapour pressure of a solution to the vapour pressure of the pure solvent.

It's a measure of how much the vapour pressure of a solvent decreases when a non-volatile solute is added.

This lowering is directly proportional to the mole fraction of solute.



[Fig: Lowering of Vapour Pressure on adding non-volatile solute]

$$(p^\circ - p) / p^\circ = \chi_B = n_B / (n_A + n_B)$$

2) Elevation of Boiling Point (ΔT_b):

The vapour pressure of a liquid increases with increase of temperature. It boils at the temperature at which its vapour pressure is equal to the atmospheric pressure.

Vapour pressure of the solvent decreases in the presence of non-volatile solute.

The boiling point of a solution is always higher than that of the boiling point of the pure solvent. ($T_2 > T_1$)

$$\Delta T_b = K_b \cdot m = (K_b \cdot W_B) / (M_B \cdot W_A)$$

where K_b = elevation constant (S.I. unit = K kg mol^{-1})

3) Depression of Freezing Point (ΔT_f):

Depression of freezing point, a colligative property, refers to the decrease in the freezing point of a solvent when a non-volatile solute is added.

This happens because the solute particles interfere with the solvent molecules ability to freeze into a solid lattice, requiring a lower temperature to reach the freezing point.

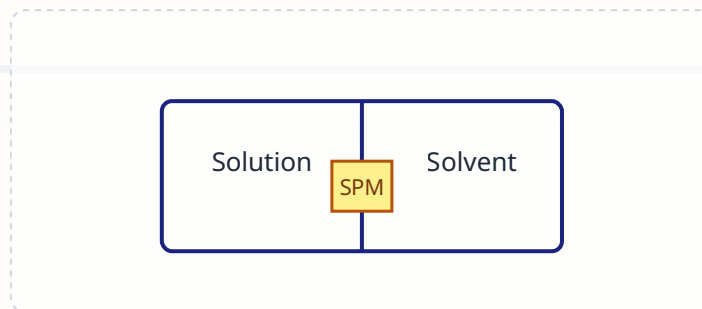
$$\Delta T_f = T_1 - T_2 = K_f \cdot m = (K_f \cdot W_B) / (M_B \cdot W_A)$$

where K_f = depression constant (S.I. unit = K kg mol^{-1})

Osmosis and Osmotic Pressure:

Osmosis: Movement of water (solvent) molecules from higher conc. of solvent to lower conc. of solvent through selective permeable membrane (or lower conc. of solute to higher conc. of solute through S.P.M.).

Osmotic Pressure (π): Pressure applied in opp. direction to stop osmosis.



[Fig: Osmosis Apparatus with SPM]

$$\pi = C \mathcal{R} T = (n / \mathcal{V}) \mathcal{R} T \Rightarrow \pi \mathcal{V} = n \mathcal{R} T = (W_B / M_B) \mathcal{R} T$$

where $\mathcal{R} = 0.0821 \text{ L atm } \mathcal{K}^{-1} \text{ mol}^{-1}$ or $8.314 \text{ J } \mathcal{K}^{-1} \text{ mol}^{-1}$

~ **Reverse Osmosis (RO):**

The direction of osmosis can be reversed if a pressure larger than the osmotic pressure is applied to the solution side. The pure solvent flows out of the solution through the semi-permeable membrane. This phenomenon is called reverse osmosis.

Solubility & Henry's Law:

Solubility: Maximum amount of solute which can be dissolved in given amount of solvent at given temperature (Solubility = $\max^m$ concentration).

Factors affecting solubility of a solute in a liquid:

1. **Nature of solute:** Ionic solutes having high lattice energy are less soluble. Hydration energy released makes ionic solids more soluble.
2. **Nature of solvent:** Ionic solids dissolve to a larger extent in a solvent having a high dielectric constant.
3. **Pressure:** Pressure does not have any significant effect on solubility of solids in liquids because they are highly incompressible. Solubility of gas in liquid $\propto$ Pressure.
4. **Temperature:** Solubility of solid in liquid $\propto$ Temperature (for endothermic).

Henry's Law:

The mass of a gas dissolved by a given volume of liquid, at constant temperature is proportional to the pressure of a gas.

$$m \propto p \Rightarrow m = K_p$$

$p = K_H \cdot x_B$ (Partial pressure in vapour phase is proportional to mole fraction)

Temp $\uparrow \Rightarrow$ Solubility $\downarrow \Rightarrow K_H \uparrow$

Application of Henry's Law:

1. *Production of Carbonated Soft Drinks:* To increase the solubility of CO_2 in soft drinks and soda water, the bottles are sealed under high pressure.
2. *Scuba Diving (The Bends):* Scuba divers breathe compressed air at high depth. Increased pressure causes nitrogen to dissolve more in blood. When they surface, pressure decreases, releasing nitrogen bubbles which block capillaries and cause a painful/dangerous medical condition called the bends. To avoid this, tanks are diluted with helium (11.6% helium, 56.2% nitrogen, 32.2% oxygen).
3. *High Altitude Respiration (Anoxia):* At high altitudes, the partial pressure of oxygen is low, leading to low concentration of oxygen in the blood and tissues of climbers, causing weakness and inability to think clearly (a condition known as anoxia).

Raoult's Law:

For a solution of volatile liquids, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction.

$$p_A = p_A^\circ x_A \quad | \quad p_B = p_B^\circ x_B$$

$$p_{\text{total}} = p_A + p_B = p_A^\circ x_A + p_B^\circ (1 - x_A)$$

Ideal & Non-Ideal Solutions:

Ideal solutions: Solutions which follow Raoult's law i.e. $p_t = p_A + p_B$. ($\Delta H = 0$, $\Delta V = 0$).

Non-ideal solutions: Which do not follow Raoult's law i.e. $p_t \neq p_A + p_B$.

- *Positive deviation:* $p_t > p_A + p_B$, $\Delta H = +ve$, $\Delta V = +ve$ (e.g., Acetone & Benzene).
- *Negative deviation:* $p_t < p_A + p_B$, $\Delta H = -ve$, $\Delta V = -ve$ (e.g., Phenol & Aniline).

Azeotropic mixtures:

- A mixture of two liquids which boils at a constant temperature and distils over without any change in composition is called Azeotropic mixture.
- Azeotropic mixtures which boils at a lower temperature than either of two components is known as azeotropic mixture with minimum boiling point. This is formed by non-ideal solution showing positive deviation. E.g.: ethanol - water.
- Azeotropic mixtures which boils at higher temperature than either of the two pure components is known as azeotropic mixture with maximum boiling point. This is formed by non-ideal solution showing negative deviation. E.g.: 68% nitric acid and 32% water by mass with a boiling point of 393.5 K.

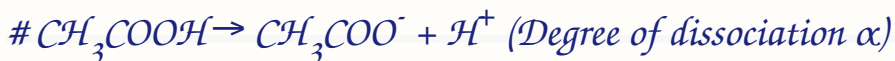
Van't Hoff's factor (i):

$$i = (\text{no. of particles after dissociation / association}) / (\text{no. of particles before dissociation / association})$$

Modified Colligative Equations:

- $\Delta T_f = i K_f m$
- $\Delta T_b = i K_b m$
- $\pi = i C R T$

Examples:



$$i = 1 + \alpha \quad | \quad K_a = [\alpha^2 / (1 - \alpha)] C \text{ (at Equilibrium)}$$